14 Show that $\dfrac{2x^2 + 10x - 48}{x^2 - 64}$ can be written in the form $\dfrac{ax + b}{cx + d}$ where a, b, c and d are integers.

$x^2 - 64 = (x + 8)(x - 8)$

1 mark
Difference of two squares.

$2x^2 + 10x - 48 = 2(x^2 + 5x - 24)$

$= 2(x + 8)(x - 3)$

1 mark
Factor the numerator.

$$\frac{2(x + 8)(x - 3)}{(x + 8)(x - 8)} = \frac{2x - 6}{x - 8}$$

1 mark
Cancel common x + 8.

Valid for $x \neq -8$ and $x \neq 8$.

(Total for Question 14 is 3 marks)

THIS AREA DO NOT WRITE IN THIS AREA